



Mitochondrial DNA structure of the Iberian populations of the white-clawed crayfish, *Austropotamobius italicus italicus* (Faxon, 1914)

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ABSTRACT

This study describes the mitochondrial structure of the Iberian population of the white-clawed crayfish *Austropotamobius italicus* under a phylogeographic context and considering the genetic variation of the entire taxon distributed in the southern part of Europe. Current sub-specific hypotheses were only partially supported, and a genetic structure related to climatic events is postulated as the main factor explaining its variation. Due to a more extensive sampling and sequencing strategy, the two different mitochondrial lineages of the taxon were found in the Iberian Peninsula, and haplotype variation of populations was observed in a considerable extent regarding that reported in previous studies from other parts of its distribution range. This evidence, together with the timing of Pleistocene expansion events for the Iberian populations ($\sim 10,772$ – $20,004$ YBP) as well as an estimated splitting time from Italian populations previous to the last glacial maximum ($T = 34,858$ years) suggest that the presence of *A. italicus* in the Iberian Peninsula could be explained by a combination of contrasting dispersal histories and a complex biogeographic scenario. We propose that although possible, human translocations may not be the only cause to explain the complex distribution patterns of species strongly affected by recent climatic events.

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1. Introduction

Historical climate changes are thought to be one of the main factors that have contributed to shaping species' distribution. Indeed, the latitudinal and altitudinal species-range shifts associated with such oscillations are thought to be one of the key factors hypothesized behind evolutionary change (Hewitt, 2000, 1999, 2004). Nevertheless, the importance of Pleistocene climatic changes ($\sim 1.8 \times 10^6$ – $10,000$ years) in driving genetic divergence and speciation remains controversial. These processes have been shown to be involved in the diversification of some taxa (e.g. Barraclough and Vogler, 2002; Johnson and Cicero, 2004; Knowles, 2001; Ribera and Vogler, 2004), whereas they have had no effect for others (e.g. Kadereit et al., 2004; Klicka and Zink, 1997, 1999). The degree to which Pleistocene climate could influence the splitting of species is likely to depend on the organism's life history, ecological opportunity and regional geomorphologic history (e.g. Weir, 2006). It is therefore reasonable to expect that certain taxa

and/or geographic areas might be more likely to show increased levels of Pleistocene diversification than others.

The European white-clawed crayfish is constituted by the species complex of *A. pallipes* s. lato (Lereboullet, 1858), widely distributed in continental Europe, ranging from the Dinaric Alps to the Iberian Peninsula and the British Isles (Grandjean and Souty-Grosset, 2000). This species presents a high genetic and morphological heterogeneity, and is therefore an interesting example for studying the relative effects of Pleistocene climatic changes and glacial refugia on genetic differentiation (Santucci et al., 1997).

The high intraspecific variability across its distribution area has meant that, this group has undergone numerous taxonomic revisions over the last five decades in an attempt to classify the morphologically and genetically distinct groups identified to date (Bott, 1950; Karaman, 1963; Starobogatov, 1995). This issue has proved to be quite complex due to the unclear or even misleading morphological differences. Genetic studies suggest that *A. pallipes* should be subdivided into two different species (Grandjean et al., 2002b, 2000), being *A. pallipes* mainly distributed across most river basins from France, the Rhine and Rhone drainages of Switzerland, the southwestern tip of Germany, and the British Isles (Grandjean et al., 2002b; Souty-Grosset et al., 2006), while *Austropotamobius italicus* (Faxon, 1914), is associated to a southern Europe distribution and can be found across Italy, the Adriatic basins in Dalmatia and the Iberian Peninsula (see Fig. 1) (Fratini et al., 2005;

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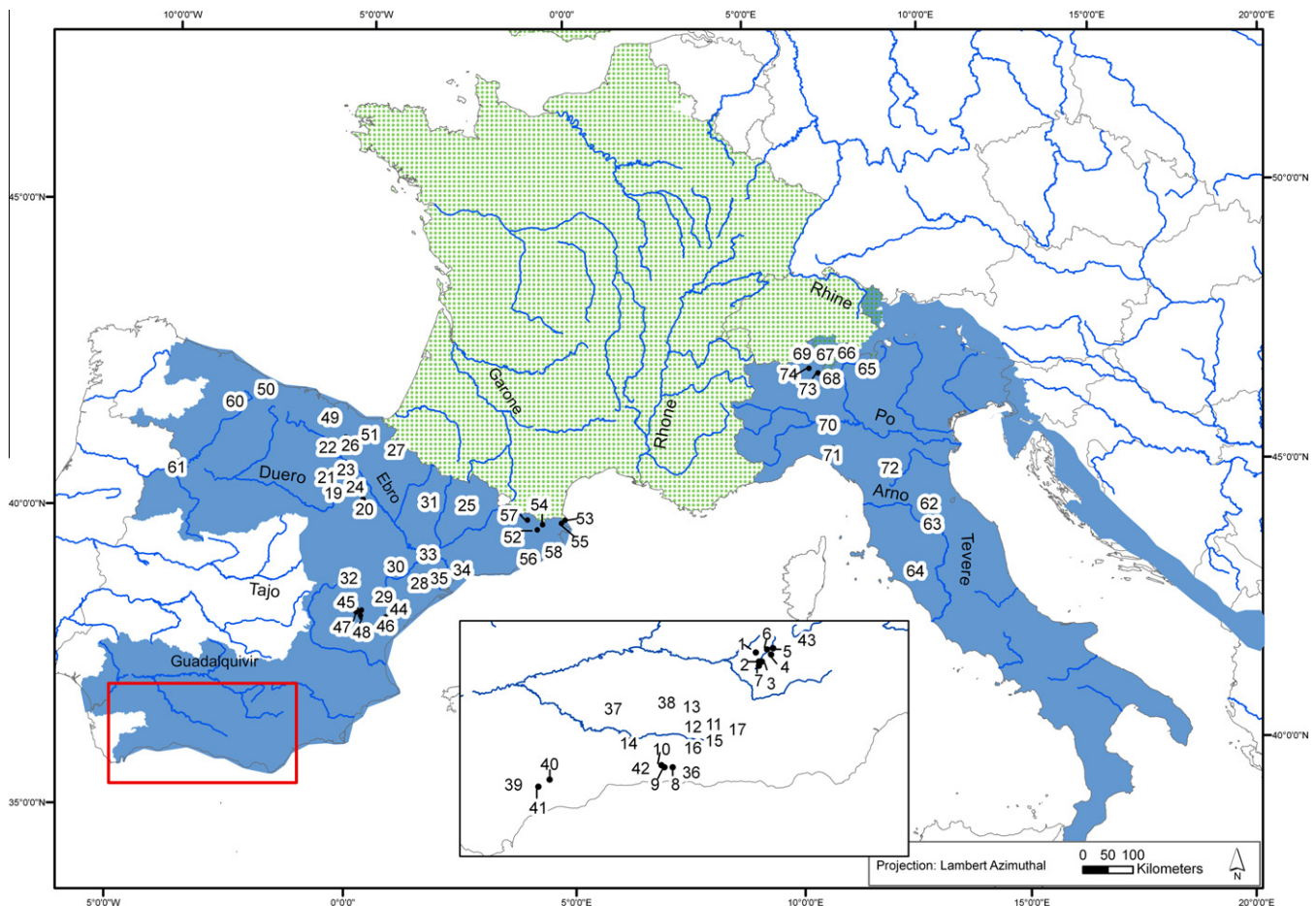


Fig. 1. Map of localities sampled in this study. Numbers correspond to localities showed in Table 1. Background patterns indicate approximate distribution ranges, as indicated by the presence of at least one population in a given watershed, although real distribution is much more restricted; solid area, distribution of *A. italicus*; squared area, distribution of *A. pallipes* (modified from Baric et al. (2005), Grandjean et al. (2002b), Souty-Grosset et al. (2006), Trontelj et al. (2005)).

Grandjean et al., 2000; Souty-Grosset et al., 2006). As a result of its wider acceptance, this will be the taxonomic hypothesis used in the present paper.

The taxonomic status of *A. italicus* is currently based on sequences from the 16S rRNA mitochondrial gene (Grandjean et al., 2002a, 2000), partially supported by morphological (Bott, 1950; Brodsky, 1983; Grandjean et al., 1998; Karaman, 1962) and allozyme data (Santucci et al., 1997). Four sub-species are accepted: *A. i. carsicus* M. Karaman, 1962 *A. i. carinthiacus* Albrecht, 1981, *A. i. italicus* (Faxon, 1914) and *A. i. meridionalis* (Fratini et al., 2005). The sub-species *A. i. italicus* is mainly found in the Central Apennines in Italy, at two localities at the South of France and Spain, *A. i. carsicus* is distributed across the Dinaric and Eastern Alps, *A. i. carinthiacus* is found in Central and North Western Italy, Austria and Switzerland and the sub-species *A. i. meridionalis* is mainly found in Southern Italy and Slovenia (Fratini et al., 2005).

The patchy and complex distribution pattern of *A. italicus* does not, however, always match the ranges proposed for its sub-species, as shown by different molecular studies where the genetic structure did not correspond to the geographic origin of the samples (Grandjean et al., 2000; Trontelj et al., 2005). The latter works suggest that the genetic patterns identified to date are not able to explain the geographical distribution of the sub-species and consequently fail to provide a well-supported framework to interpret the evolutionary history of this species.

The evolutionary history and origin of the *A. italicus* populations in the Iberian Peninsula are particularly unclear and somewhat controversial. Thus, the first molecular studies based on mitochon-

drial data found very low genetic diversity within the Iberian populations of *A. italicus* as well as shared haplotypes and non-significant differentiation between the Iberian sub-species *A. i. lusitanicus* (Bott, 1950) and the Italian *A. i. italicus* (Grandjean et al., 2002a, 2000; Iaconelli, 2001; Santucci et al., 1997; Trontelj et al., 2005). These observations led researchers to conclude that the Iberian populations of white-clawed crayfish were actually *A. i. italicus* introduced from northwestern Italy (Grandjean et al., 2002a, 2000; Trontelj et al., 2005).

However, subsequent studies (Beroiz et al., 2008; Dieguez-Urbeondo et al., 2008), found a higher genetic variation in the Iberian Peninsula and at least two different mtDNA haplotypes with certain geographical congruence, suggesting that *A. italicus* in the Iberian Peninsula had an autochthonous origin.

The debate is therefore still open, and more exhaustive studies and robust databases are needed to provide convincing explanations for the phylogeographic structure and history of *A. italicus*, particularly when complex and recent patterns are involved (Schmitt et al., 2006), as could be the case for the Iberian populations. Our aim here is to propose a valid phylogeographic scenario that would explain the evolution and origin of the Iberian populations of *A. italicus*.

Two different scenarios, each of which would result in contrasting genetic features, could be proposed based on the natural or human-mediated origin of Iberian populations of *A. i. italicus*. Thus, if *A. italicus* was recently introduced into the Iberian Peninsula from Italy, it would be expected that most of the haplotypes between *A. i. italicus* from Italy and the Iberian Peninsula would

be shared, and that there would be a lower nucleotide diversity in the latter, due to a founder effect. However, if *A. italicus* were an autochthonous species in the Iberian Peninsula, we would expect a certain concordance between geography and genetic structure as a consequence of the degree of isolation, which could exist between geographic regions due to the time since separation of population groups, even if these are recent. Furthermore, we would expect to obtain higher divergence estimates between Iberian and Italian populations of this species in the latter scenario than in the former.

To test these hypotheses, we gathered the widest distributional and molecular data set for *A. italicus* in the Iberian Peninsula to date. These data were compared with all the homologous DNA sequences available previously and analyzed at the phylogenetic and population levels.

2. Materials and methods

2.1. Tissue and data collection

Two hundred and four specimens of *A. italicus* from 53 different populations were collected throughout most of its distribution range in the Iberian Peninsula (Fig. 1). Moreover, 19 individuals from three Italian populations attributed to three different subspecies: *A. i. italicus*, *A. i. carinthiacus* and *A. i. carsicus*, were also sequenced and included in this study (Table 1). Initially, samples from the Iberian Peninsula were assigned to *A. i. italicus* following Grandjean et al. (2000). Tissue samples collected included claws or legs that can be regenerated. Individuals were captured by hand or using small aquarium nets, and then returned to the place of capture immediately after removing the tissue sample, which was placed in a vial containing 95% ethanol. All collected samples are stored at the Museo Nacional de Ciencias Naturales, CSIC, Madrid, Spain. Genomic DNA was extracted using standard methods of phenol:dichloromethane-isoamyl alcohol (24:24:1) and ethanol precipitation method (Sambrook et al., 1989).

Previous works have not always analyzed the same mitochondrial genes. Thus, to be able to perform the widest comparison possible we designed the following strategies (for reference to data sets see Fig. 2 and Table 1). (1) Data set I included a fragment of the 16S rDNA together with 45 homologous sequences available in GenBank for *A. italicus* as well as three sequences of *A. pallipes* and two of *A. torrentium* that were included as outgroups ($n = 252$ samples). This data set was used in order to allocate our samples under the phylogenetic scenario proposed by previous authors (Fratini et al., 2005; Grandjean et al., 2002a, 2000; Largiadere et al., 2000). (2) Data set II included 657 bp of the 5' portion of Cytochrome Oxidase subunit I (COI) of 241 specimens of *A. italicus* (18 sequences were downloaded from GenBank). This data set was used to infer population genetics parameters i.e. haplotype identification and genetic diversity indexes. (3) Data set III was used to infer historical population processes and dating, and was constructed using a subset of data set II (containing only sequences assigned to *A. i. italicus* of the COI gene 211 samples, 657 bp). This data set was used to apply the Isolation-Mutation Model (IM) and the demography estimation using a Bayesian Skyline Plot (BSP) and mismatch analyses. (3) Data set IV included two fragments of the COI gene (1196 bp) and the 16S rDNA fragment from all our samples were used together in a Nested Clade Analysis (NCA) to test for possible historical patterns from population structure and geographical distribution (197 samples, 1720 bp).

COI was amplified in two fragments using primers ORCO1F (5'-AACGCAACGATGATTTTTTCTAC-3'; (Taylor and Hardman, 2002) and ApaCOI1H (5'-ATAGCGACTATAGCATAAATTATC-3'; this study) for the 5' region and C1-J-2183 (5'-CAACATTTATTTTGATTTTT

GG-3'; (Simon et al., 1994) and COI *Scylla* (5'-TTAAGTCCTA GAAAATGTTGRGGGA-3'; (Gopurenko et al., 1999) for the 3' fragment. Selective amplification of 16S rDNA was also carried out with primers 16S-1471 and 16S-1472 (Crandall and Fitzpatrick, 1996). PCR amplifications were carried out in 25 μ L reactions containing: 1 $\times$ PCR buffer, 0.5 μ M of each primer, 0.2 mM of each dNTP, 1.5 mM $MgCl_2$, 1 U *Taq* polymerase (Biotools) and about 50 ng of template DNA. Cycling profile for PCR amplifications was 3 min at 94 °C (1 cycle), 30 s at 94 °C, 30 s at 48 °C and 60 s at 72 °C (30 cycles), followed by a final extension of 4 min at 72 °C. PCR products were visualized in 1.0% agarose gels (1 $\times$ TBE) stained with SYBR-Safe (Invitrogen). Fragments were sequenced on an ABI 3730 DNA Analyzer. Partial sequences of the COI and 16S rRNA genes were aligned using the computer program ClustalW (Thompson et al., 1994) as implemented in MEGA version 4 (Tamura et al., 2007). In the case of the COI gene, recommendations to detect the occurrence of possible nmtDNA were carried out for each sequence. These included the identification of stop codons and unusual levels of genetic divergence and non synonymous substitution in samples from the same population (Buhay, 2009).

2.2. Phylogenetic analyses

To infer the most complete phylogenetic framework as possible we analyzed 526 bp of the 16S rRNA gene from 252 samples of *A. italicus*. We estimated phylogeny with this mitochondrial fragment alone because it is from this that we have the widest representation of the genetic variation available from previous studies and because no data combination was accurate due to non-overlapping of data sets from GenBank. Analyses including more genes resulted in a high proportion of missing data, which led to artefactual groupings (data not shown). In order to identify the most appropriate evolutionary model by the Akaike corrected information criterion (AICc), the program jModeltest v0.1.1 (Posada, 2008) was used. Phylogenetic analyses were conducted using Maximum Likelihood with the computer program PhyML (Guindon and Gascuel, 2003). We assessed confidence in branches using 1000 non-parametric bootstrap replicates under the most accurate evolutionary model. This model was also used to carry out Bayesian inference of phylogeny as implemented in MrBayes v3.1.2 (Ronquist and Huelsenbeck, 2003), applying a Monte Carlo Markov Chain (MCMC) procedure after 5×10^6 generations. Convergence between the different run parameters in paired simultaneous runs (4 chains by run), and run length was adjusted considering an adequate sampling based on average standard deviation of split frequencies being <0.01 (Huelsenbeck and Ronquist, 2005). The burn-in period was determined as the set of trees saved prior to log likelihood stabilization and convergence as estimated using Tracer v1.5 (Rambaut and Drummond, 2007). Tracer v1.5 was used to check for convergence between chain runs and optimal values of run parameters.

2.3. Genetic diversity and demographic history

In order to evaluate genetic diversity parameters we used 657 bp of the 5' fragment of COI (Data set IV, Table 1) for 241 samples of Iberian and Italian *A. i. italicus*. This fragment was used in order to make results comparable with other studies regarding the genetic estimates from other regions for *A. italicus* (Trontelj et al., 2005), while for the COI3' fragment the previous work only had sequences from the Po basin (Zaccara et al., 2004). Geographical regions in the Iberian Peninsula were determined according to the river basins sampled (see Table 1). DnaSP v5 (Librado and Rozas, 2009) was used to generate a haplotype distribution according to the proposed regionalization. Also, for each geographical region considered, mitochondrial haplotype diversity (H_d), and

Table 1
Sample information of analyzed *Austrotamias italicus* populations. Sample sizes for each population and for different analyses are indicated between brackets. Taxa designations is showed according to Fratini et al. (2005).

Point/label	Locality	Region (Basin)	n	Phylogeny (n) data set I 16S	Diversity data set II COI5' haplotype (n)	IM and BSP data set III COI5' (n)	NCA data set IV haplotype (n) 16S + COI5' + COI3'	SAMOVA k	Taxon
1	Fresnedilla, Jaen, ES	Guadalquivir	3	Clade I (3)	3(2)7(1)	3	I(2)II(1)	2	<i>A. i. italicus</i>
2	Vadillo, Jaen, ES	Guadalquivir	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
3	Linarejos, Jaen, ES	Guadalquivir	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
4	Borosa, Jaen, ES	Guadalquivir	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
5	Aguamula, Jaen, ES	Guadalquivir	3	Clade I (3)	3(2)8(1)	3	I(2) XVI(1)	1	<i>A. i. italicus</i>
6	Coto Ríos, Jaen, ES	Guadalquivir	4	Clade I (4)	3(3)8(1)	4	I(3) XVI(1)	1	<i>A. i. italicus</i>
7	Cazorla, Jaen, ES	Guadalquivir	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
8	Cabañeros, Granada, ES	Guadalquivir	3	Clade I (3)	3(1)4(1)5(1)	3	VI(1)VII(1)XIV(1)	3	<i>A. i. italicus</i>
9	Almijara, Granada, ES	Guadalquivir	2	Clade I (2)	3(3)	2	I(2)	1	<i>A. i. italicus</i>
10	Arenas del Rey, Granada, ES	Guadalquivir	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
11	Sillar Baja, Granada, ES	Guadalquivir	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
12	Fraile, Granada, ES	Guadalquivir	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
13	Campotejar, Granada, ES	Guadalquivir	3	Clade I (3)	6(1)	3	I(2)II(1)	1	<i>A. i. italicus</i>
14	Loja, Granada, ES	Guadalquivir	2	Clade I (2)	3(2)	2	I(2)	1	<i>A. i. italicus</i>
15	Quéntar, Granada, ES	Guadalquivir	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
16	Gójar, Granada, ES	Guadalquivir	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
17	Arroyo Granada, ES ^a	Guadalquivir	1	–	21(1)	1	–	12	<i>A. i. italicus</i>
18	Arroyo Guadix, Granada, ES ^b	Guadalquivir	1	–	10(1)	1	–	4	<i>A. i. italicus</i>
19	Bco. de Fago, Aragón, ES	Ebro	3	Clade I (3)	3(2)8(1)	3	XVI(1)	1	<i>A. i. italicus</i>
20	Rivabellosa, La Rioja, ES	Ebro	3	Clade I (3)	3(3)	3	V(3)	1	<i>A. i. italicus</i>
21	El Rasillo, La Rioja, ES	Ebro	3	Clade I (3)	3(3)	3	IV(3)	1	<i>A. i. italicus</i>
22	Leza, La Rioja, ES	Ebro	3	Clade I (3)	3(3)	3	IV(3)	1	<i>A. i. italicus</i>
23	Rigobundo, Aragón, ES	Ebro	3	Clade I (3)	3(3)	3	IV(3)	1	<i>A. i. italicus</i>
24	Faldetores, La Rioja, ES	Ebro	3	Clade I (3)	3(3)	3	IV(3)	1	<i>A. i. italicus</i>
25	Pon deSuert, ES	Ebro	10	Clade I (10)	8(9)11(1)	10	XII(1)XV(8)XVI(1)	1	<i>A. i. italicus</i>
26	Río Zadoya, Álava, ES	Ebro	2	Clade I (2)	3(2)	2	IV(2)	1	<i>A. i. italicus</i>
27	Roncesvalles, Navarra, ES ^a	Ebro	1	–	21(1)	1	–	11	<i>A. i. italicus</i>
28	Beceite, Valderrobres, ES ^a	EbroII	1	–	21(1)	1	–	10	<i>A. i. italicus</i>
29	Asabón, Aragón, ES	EbroII	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
30	Valhondo, Aragón, ES	EbroII	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
31	Hornero, Aragón, ES	EbroII	3	Clade I (3)	8(3)	3	XVI(3)	1	<i>A. i. italicus</i>
32	Amanaderos, Aragón, ES	EbroII	3	Clade I (3)	8(3)	3	XVI(3)	1	<i>A. i. italicus</i>
33	Montsant, Catalunya, ES	EbroII	4	Clade I (4)	8(4)	4	XVI(4)	1	<i>A. i. italicus</i>
34	Molí de l'Esquirola, Catalunya, ES	EbroII	3	Clade I (3)	8(3)	3	XVI(3)	1	<i>A. i. italicus</i>
35	Ports de Beset, ES	EbroII	10	Clade I (3)	3(3)8(7)	10	I(3) XVI(7)	1	<i>A. i. italicus</i>
36	Guajar, Granada, ES	Independent (Guadlfe)	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
37	Cabra, Córdoba, ES	Independent (Guadajoz)	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
38	Valdepeñas, Jaen, ES	Independent (Guadajoz)	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
39	Montejaque, Málaga, ES	Independent (Guadiaro)	3	Clade I (3)	3(2)5(1)	3	I(2) VI(1)	1	<i>A. i. italicus</i>
40	Ronda, Málaga, ES	Independent (Guadiaro)	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
41	Yunquera, Málaga, ES	Independent (Guadalhorce)	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
42	Alcaucín, Málaga, ES	Independent (Vélez)	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
43	Vites, Jaen, ES	Independent (Segura)	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
44	Zucaina, Valencia, ES	Independent (Mijares)	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
45	Arcos de las Salinas, Valencia, ES	Independent (Turia)	3	Clade I (3)	3(3)	3	I(1) XVI(2)	1	<i>A. i. italicus</i>
46	Higuera, Valencia, ES	Independent (Mijares)	3	Clade I (3)	3(3)	3	XVI(3)	1	<i>A. i. italicus</i>
47	Titaguas, Valencia, ES	Independent (Turia)	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
48	Campo de Arriba, Valencia, ES	Independent (Turia)	3	Clade I (3)	3(3)	3	XVI(3)	1	<i>A. i. italicus</i>
49	Río Nervión, Álava, ES	Northern Spain (Nervión)	2	Clade I (2)	3(2)	2	IV(2)	1	<i>A. i. italicus</i>
50	Lake Ercina, ES	Northern Spain (Lake Ercina)	4	Clade I (4)	3(4)	4	V(4)	1	<i>A. i. italicus</i>
51	Orozco, Tolosa, ES ^a	Northern Spain (Nervión)	1	–	21(1)	1	–	11	<i>A. i. italicus</i>
52	Garrotxa, Catalunya, ES	East Pyrenees (Fluvià)	10	Clade I (10)	3(9)10(1)	10	I(6) X(3)XIII(1)	1	<i>A. i. italicus</i>

Table 1 (continued)

Point/label	Locality	Region (Basin)	n	Phylogeny (n) data set I 16S	Diversity data set II COI5' haplotype (n)	IM and BSP data set III COI5' (n)	NCA data set IV haplotype (n) 16S + COI5' + COI3'	SAMOVA k	Taxon
53	Albera, Catalunya, ES	East Pyrenees (Muga)	10	Clade I (10)	3(8)12(1)13(1)	10	I(8) XX(1)XXI(1)	1	<i>A. i. italicus</i>
54	Salines, Catalunya, ES	East Pyrenees (Muga)	21	Clade II	16(16)17(5)	–	–	1	<i>A. i. italicus</i>
55	Sant Genís, Figueres, ES ^b	East Pyrenees (Muga)	1	–	10(1)	1	–	5	<i>A. i. italicus</i>
56	Fondrats, ES	East Pyrenees (Besós)	4	Clade I (4)	9(1)14(2)15(1)	4	VIII(2)IX(1)XI(1)	1	<i>A. i. italicus</i>
57	Riudaria, Catalunya, ES	East Pyrenees (Ter)	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
58	Les Preses, Catalunya, ES	East Pyrenees (Fluviá)	3	Clade I (3)	3(3)	3	I(3)	1	<i>A. i. italicus</i>
59	Pujol Avencó, Catalunya, ES	East Pyrenees (Besós)	2	Clade I (2)	3(2)	2	I(2)	1	<i>A. i. italicus</i>
60	Redipollos, Riano, León, ES ^a	Duero	1	–	21(1)	1	–	7	<i>A. i. italicus</i>
61	Tortulhas, Miranda do Douro, PT ^b	Duero	1	–	10(1)	1	–	6	<i>A. i. italicus</i>
62	Torrente Sovara, Arezzo, IT	Tevere	10	Clade I (10)	19(10)	10	XVII(2)XVIII(4) IXX(4)	13	<i>A. i. italicus</i>
63	Papiano, Arezzo, IT ^a	Tevere	1	–	21(1)	1	–	12	–
64	Resco Simontano, IT	Arno	4	Clade I (4)	1(2)2(2)	–	XXII(2)XXIII(2)	–	<i>A. i. carinthiacus</i>
65	Sondrio, IT	Po (Adda)	5	Clade II (5)	18(5)	–	–	9	<i>A. i. carsicus</i>
66	Claro, Bellinzona, CH ^a	Po	1	–	22(1)	1	–	9	<i>A. i. italicus</i>
67	Cugnasco, Lugano, CH ^a	Po	1	–	22(1)	1	–	9	<i>A. i. italicus</i>
68	Meride, Lugano, CH ^a	Po	1	–	22(1)	1	–	9	<i>A. i. italicus</i>
69	Montecrestese, Domodossola, IT ^a	Po (Stresa)	1	–	22(1)	1	–	9	<i>A. i. italicus</i>
70	Sagliano, Voghera, IT ^a	Po	1	–	22(1)	1	–	9	<i>A. i. italicus</i>
71	Ottone, Piacenza, IT ^a	Po	1	–	22(1)	1	–	9	<i>A. i. italicus</i>
72	Nirano, Sassuolo, IT ^a	Po	1	–	21(1)	1	–	12	<i>A. i. italicus</i>
73	Casalzuigno, Varese, IT ^a	Po (Ticino)	1	–	22(1)	1	–	9	<i>A. i. italicus</i>
74	Bracchio, Verbania, IT ^a	Po (Ticino)	1	–	20(1)	1	–	8	<i>A. i. italicus</i>
A29, A30	Bologne, IT ^c	Po	2	Clade I (1)	–	–	–	–	<i>A. i. italicus</i>
A31	Las Lilas, FR ^c	–	1	Clade I (1)	–	–	–	–	<i>A. i. italicus</i>
A35	Las Lilas, FR ^c	–	1	Clade II (1)	–	–	–	–	<i>A. i. meridionalis</i>
ESP	San Esteban, ES ^c	Cantabric	1	Clade I (1)	–	–	–	–	<i>A. i. italicus</i>
A32	Garrel, FR ^c	Garone	1	Clade I (1)	–	–	–	–	<i>A. i. italicus</i>
ESP	Tafalla, ES ^c	Ebro	1	Clade I (1)	–	–	–	–	<i>A. i. italicus</i>
A33, A34	Plansee, AT ^c	Danube	2	Clade I (1)	–	–	–	–	<i>A. i. carinthiacus</i>
A28	Rizana, SL ^c	Rizana	1	Clade II (1)	–	–	–	–	<i>A. i. meridionalis</i>
ESP	Navarra, ES	–	1	Clade I (1)	–	–	–	–	<i>A. i. italicus</i>
S4	Cugnasco, IT ^d	Po	1	Clade I (1)	–	–	–	–	<i>A. i. carinthiacus</i>
N1	Eyrieux/Doux, Lamastre, FR ^d	Rhone	1	Clade I (1)	–	–	–	–	<i>A. i. carinthiacus</i>
SW1	Lac Grond, Laax, GR ^d	Po	1	Clade I (1)	–	–	–	–	<i>A. i. carinthiacus</i>
SW2	Sous le Cex, Massongez, VS ^d	Po	1	Clade I (1)	–	–	–	–	<i>A. i. carinthiacus</i>
V1, V2	Monti Berici, Vicenza, IT ^d	Adige	2	Clade II (1)	–	–	–	–	<i>A. i. carsicus</i>
A3	Modolena, IT ^d	Po	1	Clade I (1)	–	–	–	–	<i>A. i. carinthiacus</i>
Bs	Chiese, Brescia, IT ^e	Po	1	Clade II (1)	–	–	–	–	<i>A. i. carsicus</i>
VAR	Throughout Po watershed ^e	Po	1	Clade I (1)	–	–	–	–	<i>A. i. carinthiacus</i>
Ard	Ardversa, Rovescala, IT ^e	Po	1	Clade I (1)	–	–	–	–	<i>A. i. carinthiacus</i>
Ver	Versa, Rovescala, IT ^e	Po	1	Clade I (1)	–	–	–	–	<i>A. i. carinthiacus</i>
Vis	Bormida, Visone, IT ^e	Po	1	Clade II (1)	–	–	–	–	<i>A. i. meridionalis</i>
A1	*f	Po, Magra	1	Clade I (1)	–	–	–	–	<i>A. i. carinthiacus</i>
A2	Lambro, IT ^f	Po	1	Clade I (1)	–	–	–	–	<i>A. i. carinthiacus</i>
A3, A10	*f	Po, Magra	2	Clade I (1)	–	–	–	–	<i>A. i. carinthiacus</i>
A4	Rosandra, IT ^f	Rosandra	1	Clade II (1)	–	–	–	–	<i>A. i. carsicus</i>
A6, A7	Farfereta, IT ^f	Arno	2	Clade I (1)	–	–	–	–	<i>A. i. italicus</i>
A8	*f	Bidente-Ronco, Arno	1	Clade I (1)	–	–	–	–	<i>A. i. italicus</i>
A9	Duranna, IT ^f	Tevere	1	Clade II (1)	–	–	–	–	<i>A. i. meridionalis</i>
A13	*f	Po, Crati, Sele	1	Clade II (1)	–	–	–	–	<i>A. i. meridionalis</i>
A14	*f	Tevere, Sele	1	Clade II (1)	–	–	–	–	<i>A. i. meridionalis</i>
A15, A16, A17	Lambro, IT ^f	Po	3	Clade II (1)	–	–	–	–	<i>A. i. carsicus</i>
A18	Lake Caldonazzo, IT ^f	Brenta	1	Clade II (1)	–	–	–	–	<i>A. i. carsicus</i>
SS2	Gravi, Lemme, IT ^e	Po	1	–	–	–	–	–	<i>A. pallipes</i>
Ap2	Gace, FR ^c	Rhone	1	–	–	–	–	–	<i>A. pallipes</i>
Ap3	Genes, FR ^c	**	1	–	–	–	–	–	<i>A. pallipes</i>
At1	– ^c	**	1	–	–	–	–	–	<i>A. torrentium</i>
At2	– ^g	**	1	–	–	–	–	–	<i>A. torrentium</i>

Superscripts refer to sequences used from previous studies: ^aIaconelli (2001), only COI 5' fragment available; ^bTrontelj et al. (2005), only COI 5' fragment available; ^cGrandjean et al. (2000) (16S); ^dLargiadet et al. (2000); ^eZaccara et al. (2004); ^fFratini et al. (2005); ^gSchubart and Huber (2006). Sequence from Navarra uploaded by Berioz et al. Countries named by the ISO code: IT Italy, PT Portugal, ES Spain, CH Switzerland. *Samples from several localities in one region; **samples coming from several regions.

Data set	Analyses	bp(n)	Included
I 	Phylogeny	524 (252)	All <i>A. italicus</i> sub-species
II 	Genetic diversity	657 (241)	<i>A. italicus</i> IT+ES
III 	Demography	657 (211)	<i>A. i. italicus</i> lineage IT+ES
IV   	NCA	1720 (197)	<i>A. italicus</i> IT+ES, complete mt sequences
16S COI5' COI3'			

Fig. 2. Scheme depicting data set used in this study. The use of the different data sets was decided according to the possibility and pertinence for different analysis, and in order to recover the widest genetic variation from other studies. Data set I was composed of 16S sequences, used in Phylogenetic analysis because it had the widest geographical covering and to locate our samples in the taxonomical background from previous studies. Data set II was composed by a 5' fragment of COI for genetic parameter estimations and population comparisons, as it provided more accurate variation to study population processes and allow comparison with previous studies. Data set III was more reduced than latter, as it was limited to describe variability in *A. i. italicus* sub-species for demography and timing. Data set IV included the complete data obtained by us to describe Nested Clade Analysis with the widest mitochondrial evidence for the Iberian Peninsula.

nucleotide diversity (π) were calculated in DnaSP (Librado and Rozas, 2009). As very different sampling sizes were available for Iberian and Italian populations, we carried out a rarefaction procedure to evaluate possible effects of this on haplotype richness as increasing sampling number, with the program EstimateS (Colwell, 2009). Chao1 non-parametric richness estimator was used to estimate the influence of private haplotypes on accumulative sampling localities (Chao, 1987).

There is evidence that climatic changes have been decisive for the distribution and recent population history of several different freshwater species in Europe. In order to detect signatures of demographic changes or selection in the recent history of *A. i. italicus*, deviations from a model of mutation-drift equilibrium were tested using the 5' fragment of COI from the Iberian and Italian populations only considering those assigned to *A. i. italicus* (data set III, $n = 191$ and 657 bp, Table 1). This fragment was used to include all closer haplotypes reported before (Trontelj et al., 2005), while the COI3' fragment was not used because sequences available were restricted to the Po river basin (Zaccara et al., 2004). We used the two most powerful tests to detect deviations from neutrality due to population expansion: Fu's F_S (Fu, 1997) and R_2 (Ramos-Onsins and Rozas, 2002) tests of neutrality (using 1000 pseudo-replications in DnaSP). Fu's F_S is especially sensitive to an excess of rare haplotypes (Fu, 1997; Ramos-Onsins and Rozas, 2002), for which it takes on low or negative values with the excess of singleton mutations caused by population expansions. R_2 uses information from segregating sites, and is more powerful for small number of segregating sites (Ramos-Onsins and Rozas, 2002). Neutrality was also assessed by the probability of obtaining raggedness values of r less than the observed, calculated using the coalescent algorithm in DnaSP over 1000 pseudo-replications with a random seed and no recombination. Tajima's D (Tajima, 1989) and F^* of Fu and Li (Fu and Li, 1993) tests of neutrality were also used to assess changes in demographic history.

Additionally, using the same nucleotide matrix, we performed a mismatch analysis, which plots the distribution of the number of differences between pairs of sequences, in DnaSP. Population expansion would appear as a "wave" in the mismatch distribution, while stable population sizes produce ragged multi-modal distributions (Harpending, 1994; Rogers and Harpending, 1992). We used initial values of $\theta_0 = 0$, and $\theta_1 = 99,999$. To obtain an approximate time corresponding with a possible expansion event leading to actual Iberian populations of *A. i. italicus* we used the expansion parameter $\tau = 2ut$ obtained in the analysis, where u is the divergence rate per nucleotide and year and t is the time in generations since expansion. The τ parameter is an estimate of time after expansion (t) in mutational units. Thus, if u ($u = 2\mu$, where μ is the substitution rate per lineage) and the number of nucleotides of the fragment analyzed (l) are known, it is possible to calculate the age when the expansion occurred using the expression $\tau = 2ut$, from what we have $t = \tau/2u$ (modified from Harpending et al. (1993)).

Three COI rates reported for other crustaceans were considered: 1.66% divergence per million years (PMY) determined for grapsid crabs (Knowlton and Weigt, 1998), 2.6% PMY obtained for snapping shrimps (Knowlton et al., 1993), and 1.4% PMY (Cook et al., 2008; Knowlton and Weigt, 1998) as an approximate average divergence rate reported for crustaceans. We considered a generation time of two years for crayfish species (Hobbs, 1991) and equal sex ratios.

Finally, to reveal changes in relative genetic diversity through time (a measure of effective population size under neutral evolution), in the Iberian and Italian populations, we inferred a Bayesian skyline plot (BSP) for each data set in BEAST v1.5.3 (Drummond and Rambaut, 2007). In each case, MCMC chains were run for 5×10^7 generations. Convergence was assessed using Tracer v1.5 and uncertainty in parameter estimates reflected in values of the 95% highest probability density (HPD). Because sample sizes were biased and, consequently, effective samples sizes might be skewed, we constructed and analyzed as described above 10 additional matrices with 20 random Iberian sequences, so they could be compared with the minimum sample size of the Italian samples.

2.4. Phylogeographic analyses

We explored global genetic structure for this species using a Spatial Analysis of Molecular Variance (SAMOVA) (Dupanloup et al., 2002). This method defines k continuously homogeneous but genetically differentiated populations, where the proportion of total genetic variance (F_{CT}) is maximized. We run SAMOVA from $k = 2$ to $k = 15$ with 500 simulated annealing processes. Each run was repeated three times to check for consistency. For this analysis we used the 657 bp COI data set, including 211 samples from the Iberian and Italian Peninsula, which were grouped according to their river basin (Table 1).

As a complementary method to those performed before, and considering the limitations and drawbacks of NCA (Panchal and Beaumont, 2007; Petit, 2008), we tested geographical associations among haplotypes and clades, based on the most parsimonious haplotype network followed by a nested cladistic analysis (Templeton, 1998, 2001). Only two populations from France have been found to harbor *A. i. italicus* haplotypes, Les Liles (which had *A. palipes* also), and Garrel (Grandjean et al., 2000). For this reason, its origin by human translocation has been argued (Grandjean et al., 2000; Souty-Grosset et al., 2006). We cannot, however, rule out the possibility of losing haplotype representation from these populations, but it seems difficult that a gradient of haplotypes could exist in them. In addition, due to this sampling limitation, we used NCA to test for possible geographic associations between clades of the Iberian Peninsula and included the two closer Italian populations as a reference (populations 62 and 64). In order to account for most of the mitochondrial information, we used the data set IV ($n = 197$, 16S + COI = 1720 bp, Fig. 2, Table 1) composed of sequences obtained in this study. We first used the program TCS

v1.21 (Clement et al., 2000) to build the most parsimonious haplotype network that would illustrate the mutational step between sequences. GEODIS v2.6 (Posada et al., 2000) was then used to test for significant relationships between geographic locations and genetic distances for 10,000 random permutations.

Biological inferences for clades showing significant geographical association were drawn using the updated inference key (15 December 2008, available at http://darwin.uvigo.es/download/geodisKey_28Apr09.pdf; (Templeton, 2004)) to determine which historical processes might have most probably lead to the current patterns.

2.5. Isolation-Mutation Model

To approach the two proposed scenarios of natural colonization of the Iberian Peninsula or anthropogenic introduction from Italy we applied the Isolation-Mutation as a tool to test the hypothesis of a shared-history scenario for Iberian and Italian populations of *A. i. italicus*. Thus, we proposed a putative unique population for both Iberian and Italian populations in the past regarding an initial gene flow with posterior splitting between them, and we carried out a pairwise comparison for genealogical estimates of diversity (θ_1 , θ_2 , θ_{Ancestor}) and splitting time using the program IM (Isolation-Migration Model, Hey, 2005; Nielsen and Wakeley, 2001; Won et al., 2005; <http://lifesci.rutgers.edu/heylab/heysoftware.htm#IM>). This model has proven to be specially useful to estimate population parameters from the process of population divergence as time since splitting and population size changes (Won and Hey, 2005), which eventually could be useful to study scenarios as the one approached in this study. Providing a temporal scenario would be especially useful for the hypothesis of a unique ancestral population for Italian and Iberian populations and a posterior splitting. This model assumes that an ancestral population of constant size and population mutation parameter θA separated into two populations at time T . This is obtained as a function of the mutation rate ($t = T\mu$), from which we can estimate T in years ($T = t/\mu$). Estimates of population sizes were also converted from θ , as it is expressed in mutational units ($\theta = 4N\mu$, from which $N = \theta/4\mu$).

Each descendant population has its own mutation parameter, θ_1 and θ_2 , respectively. The comparison was made with the data set III because it included the widest range of samples from the Iberian and NW Italy haplotypes (considered here as the *A. i. italicus* lineage, 211 samples and 657 bp). Comparisons were carried out grouping together all the Iberian populations except Muga Salinas on one side and the Italian populations on the other side. The HKY (Hasegawa–Kishino–Yano) model with an inheritance scalar of 0.25 for mitochondrial DNA was used with a random seed to initiate the run. A burn-in of 100,000 steps was discarded before recording genealogical steps, and each comparison was run until all effective sample sizes (ESS) were larger than 1000. Multiple independent runs with random start seeds were performed to ensure values were converging on similar estimates. We used a rate divergence for COI of 1.4% PMY (Cook et al., 2008; Knowlton and Weigt, 1998) as an approximate to the average rate commonly observed for crustaceans.

3. Results

3.1. Phylogeny of *Austropotamobius italicus*

The 16S rDNA phylogeny was inferred to help assign the populations of this study to the different sub-species previously identified for the *A. italicus* species complex (Fratini et al., 2005; Grandjean et al., 2000). The selected model of nucleotide evolution

was HKY85 (Hasegawa et al., 1985) with a gamma distribution shape parameter of 1.35 and a proportion of invariable sites of 0.71. Log-Likelihood stabilization from BI was observed after burning a 10% of states, and tree scores were -1327.02034 and -1656.02 for the ML and BI methods, respectively. The phylogenetic analysis of *A. italicus* showed two main clades, Clade I, which contains samples attributed to *A. i. italicus* and *A. i. carinthiacus*, and Clade II, which contains samples of *A. i. meridionalis* and *A. i. carsicus* (Fig. 3). Most of the samples from Iberia included in the Phylogenetic analysis showed identical 16S rDNA haplotypes, thus making the assignation of the majority of the Iberian populations to this sub-species clear (Grandjean and Souty-Grosset, 2000). The Iberian population from the river Muga, East Pyrenees (Muga-Sal) was included in Clade II, together with those samples attributed to *A. i. meridionalis*.

3.2. Haplotype and nucleotide diversity

Haplotype analysis of the 5' fragment of COI, showed 15 haplotypes found exclusively in the Iberian Peninsula, as well as four haplotypes for the Italian populations (Table 1). Two haplotypes (Haplotypes 10 and 21) were shared between both regions, and most haplotypes were found at low frequencies with one common haplotype (Haplotype 3) being observed in 70% of the Iberian populations. Rarefaction curves showed no apparent effect of different sample sizes, as when Chao1 estimator was considered, this resulted in lower values for the Italian than for Iberian populations in the same points and the curve of the latter populations did not reach an asymptote, meaning that even more haplotypes could be expected if more populations are added.

Overall, genetic diversity values (π and H_d) were much lower for the Iberian ($\pi = 39 \pm 0.000$; $H_d = 0.121 \pm 0.032$) than for the Italian rivers ($\pi = 94 \pm 0.0006$; $H_d = 0.671 \pm 0.066$). In the Iberian Peninsula, the Ebro basin was the most diverse region considering haplotypic diversity and the Guadalquivir basin showed the lowest nucleotide and haplotypic diversity (Table 2). However, the East Pyrenees region, which contains the populations of the river Muga, was the region with the highest average number of polymorphic sites ($k = 0.911$) and also with the highest nucleotide diversity ($\pi = 0.0014 \pm 0.0002$).

Most pairwise Φ_{ST} comparisons between populations and geographic groups of *A. i. italicus* were non-significant (Table 3). The highest and significant Φ_{ST} values were observed when comparing the Muga Salinas population with the Guadalquivir and High Ebro populations ($\Phi_{ST} = 0.976$ and 0.940 , respectively). High Φ_{ST} values were also found between the Italian populations of *A. i. italicus* and those in the Iberian river basins. The SAMOVA analysis clearly separated the Iberian and Italian populations, from the lowest k values. However, the highest F_{CT} value ($F_{CT} = 0.995$) was obtained for an arrangement of the predefined populations in $k = 13$ clusters (Fig. 4, Table 1). Generally, a large genetic cluster was obtained containing 52 Iberian localities, whereas the rest of localities from the Iberian Peninsula were found mostly as eight separate single clusters. The Italian samples were organized in four genetic clusters, one of which contained seven localities. Only one genetic group included Iberian and Italian samples, as the Arroyo River from the Guadalquivir basin grouped together with the Nirano and Papiano rivers from the Po and Tevere basins, respectively (Table 1).

3.3. Nested clade phylogeographic analysis

The haplotype parsimony network provided by TCS for the total mitochondrial information was used to construct a nested design (Fig. 5), which was structured into 13 first-level clades, six second-level clades and four third-level clades. All the Iberian samples

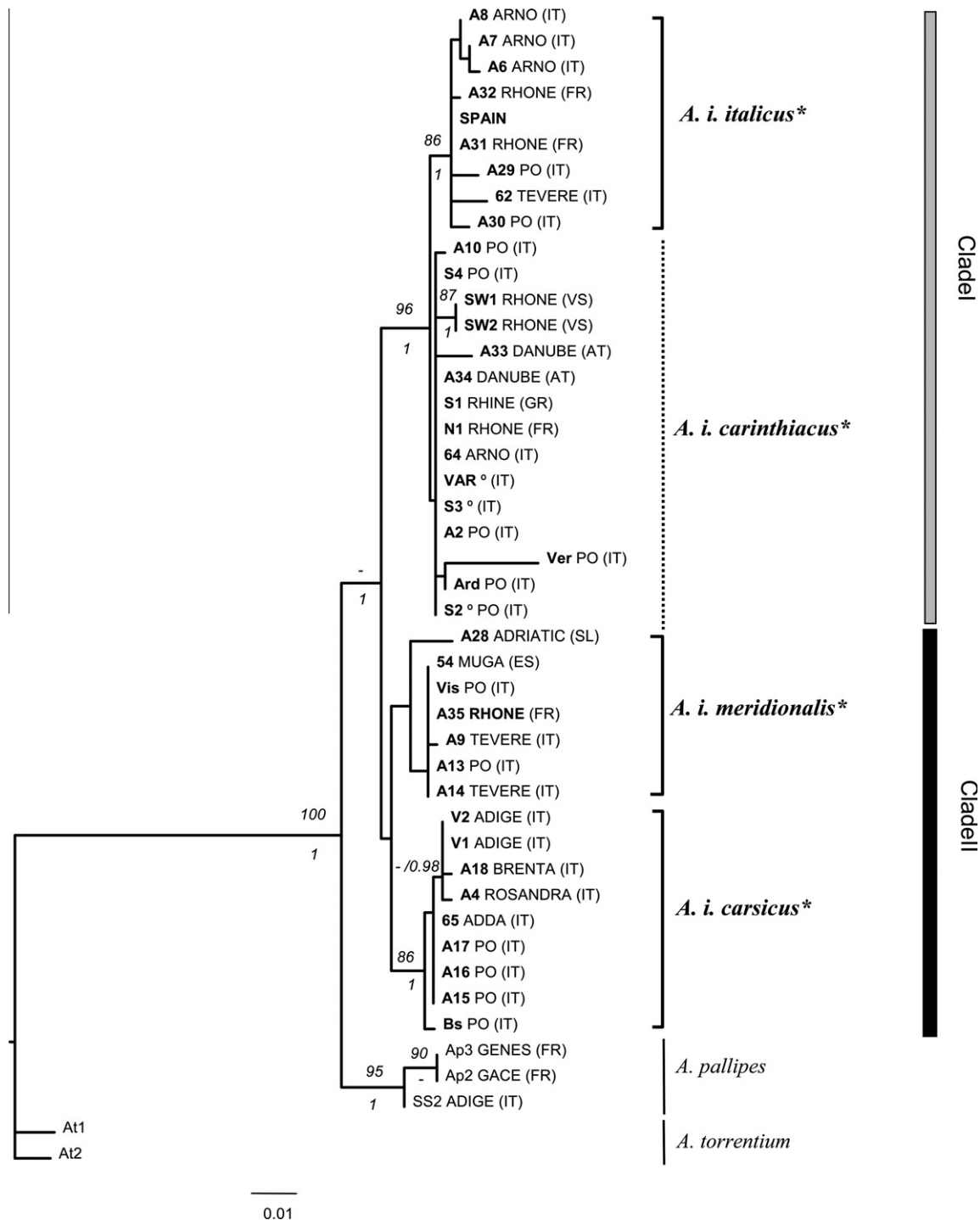


Fig. 3. Phylogenetic tree derived from the ML analysis of 16S rDNA data of the *A. italicus* populations published to date, using the model HKY+I+G. Included are three populations from Italy sequenced for this study representing three of the four recognized sub-species of *A. italicus* (Fratini et al., 2005), and 52 Iberian populations analyzed for this study (labeled as SPAIN). Bootstrap values for the ML analysis are given above nodes (1000 replicates) and posterior probability values for bayesian analysis are shown below nodes (5×10^6). Only bootstrap values higher than 80 and posterior probabilities higher than 0.95 are shown. Labels include the haplotype code for 16S rDNA (see Supplementary file 1), river system and country. ° makes reference to several river basins for one haplotype (Zaccara et al., 2004). *Refers to the taxonomy subdivision proposed by Grandjean et al. (2000) and Fratini et al. (2005).

were clustered in Clade 3-3, whereas the Italian haplotypes were found in Clades 3-1 and 3-2 (Table 4). Nine of the 20 clades with geographical and genetic information used in GeoDis, showed significant non-random associations between clades and sampling localities (Table 5). For example, Clades 2-5, 2-6 and 2-7 showed significant geographical association corresponding to restricted

gene flow with isolation by distance (Table 5). Clade 2-7 mainly comprises populations from North Eastern Iberia and the Pyrenees and Clade 2-5 exclusively contains samples from the Guadalquivir basin, whereas Clade 2-6 covers almost all the geographical distribution of the species in the Iberian Peninsula. Several single-step clades (1-7, 1-8, 1-14) showed significant associations, all of them

Table 2

Mitochondrial COI genetic diversity in different geographical areas of *Austroptamobius i. italicus*, using the data set II of the gene.

Region/group	n	h	S	H _d	k	π
<i>A. italicus</i> Italy	20	4	8	0.671 ± 0.066	3.730	0.0090 ± 0.0006
<i>A. italicus</i> Iberian	191	5	4	0.121 ± 0.032	0.162	0.0004 ± 0.0001
Spain_indep_basins	33	3	2	0.392 ± 0.087	0.405	0.0006 ± 0.0015
Guadalquivir	46	6	6	0.245 ± 0.084	0.344	0.0005 ± 0.0002
Ebro (High + Low)	60	3	2	0.524 ± 0.210	0.540	0.0008 ± 0.0000
Duero	2	1	0	0.000	0.000	0.000
NSpain	7	1	0	0.000	0.000	0.0000
East Pyrenees	33	7	6	0.383 ± 0.107	0.911	0.0014 ± 0.0002
Sondrio	5	1	0	0.000	0.000	0.0000
Tevere	10	1	0	0.000	0.000	0.0000
Arno	5	2	4	0.667 ± 0.204	2.667	0.0040 ± 0.0012

n, number of individuals; h, number of haplotypes; S, number of polymorphic sites; H_d = haplotype diversity; k = average number of polymorphic sites; π, nucleotide diversity.

corresponding to “restricted gene flow/dispersal but with some long-distance dispersal over intermediate areas not occupied by the species; or past gene flow followed by extinction of intermediate populations”.

3.4. Demography and timing

Neutrality tests applied to the data set III showed a significant deviation from neutrality in the case of Fu's $F_s = -11.27$ ($P < 0.05$) and $R_2 = 0.0299$ ($P < 0.05$), despite the other tests showed non-significant deviations from equilibrium. Tajima's D (-1.34 , $0.10 > P > 0.05$) and Fu and Li's F (-1.78 , $0.10 > P > 0.05$) were negative, although neither value nor the raggedness statistic ($Hr = 0.064$) was significant. This incongruence can be explained by the fact that the two significant tests are specially indicated to detect deviations due to population expansion. The other tests are prone to be influenced by sampling size and/or the occurrence of haplotypes at low frequencies, and therefore have less statistical power (Fu, 1997; Ramos-Onsins and Rozas, 2002).

A unimodal mismatch distribution, which is commonly inferred for populations that have undergone expansion (Harpending, 1994; Rogers and Harpending, 1992), was observed in the case of the two COI fragments analyzed and the total mitochondrial evidence from Iberian populations of *A. i. italicus* (see Supplementary file II). Given the range of mutation rates for COI, the resulting value of $\tau = 0.184$ for the Iberian samples suggests an expansion of an original population between 16,871 and 10,772 years BP for the 1.66% and 2.6% divergence rates, and about 20,004 years BP considering an average divergence rate of 1.4%. Independent runs using IM converged on approximate marginal posterior probability distributions and we were able to obtain reliable estimates for θ_1 and θ_2 . The results of the IM analysis are summarized in Fig. 6. Time since divergence ($t = T\mu$) scaled by the mutation rate (μ) was also recovered with confidence because its distribution was almost complete (although at times non-zero) after the stabilization procedure. The sharp peak in the marginal posterior probability distribution of the divergence time parameter strongly indicates a recent split between *A. i. italicus* from the Iberian and Italy. When t is converted to a scale of years for the mutation rate ($\mu = 0.7\%$, PMY, $t = 0.16$, see Fig. 6) corresponds to $T = 34,858$ years, (the lower and upper bounds of the 95% confidence interval were 20,261– 1.50×10^6 years). The position of the peaks of the marginal posterior densities (Fig. 6) for the θ parameter were $\theta = 15.02$ (9.6051–111.3293, 95%HPD) and 5.89 (2.4819–48.5091, 95%HPD) for the Iberian and Italian populations, and $\theta_A = 8.2$ (1.6922–141.7402, 95%HPD) for the ancestral inferred population. This corresponds to a population size of 818,182 for N1 (523,153– 6.06×10^6 95% HPD), 321,269 for N2 (135,179– 2.64×10^6 95%

Table 3 Pairwise Φ_{ST} comparisons between geographic groups of *Austroptamobius italicus*. Values in bold indicate statistical significance after sequential Bonferroni correction [$P(i) = 0.00028$], and those underlined indicate significant values after False Discovery Rate correction [$P(i) = 0.0097$]. Negative values are due to algorithm application, and should be regarded as zero values.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
1. Arno																			
2. Guadajoz	0.806																		
3. Guadalfeo	0.720	0.000																	
4. Guadalquivir	0.921	-0.126	-0.314																
5. Guadlario	0.755	0.111	-0.200	-0.052															
6. Guadalhorce	0.720	0.000	0.000	-0.314	-0.200														
7. Velez	0.720	0.000	0.000	-0.314	-0.200	0.000													
8. Segura	0.720	0.000	0.000	-0.314	-0.200	0.000	0.000												
9. High Ebro	0.857	0.195	0.081	0.316	0.194	0.081	0.081	0.081											
10. Low Ebro	0.927	0.539	0.485	0.427	0.540	0.485	0.485	0.485	<u>0.245</u>										
11. Mijares	0.792	0.333	0.111	0.150	0.200	0.111	0.111	0.111	0.068	-0.054									
12. Turia	<u>0.831</u>	0.314	0.143	0.182	0.250	0.143	0.143	0.143	0.112	-0.010	-0.263								
13. Nervion	0.636	1.000	1.000	0.522	0.500	1.000	1.000	1.000	-0.149	0.748	0.556	0.600							
14. Erquina	0.751	1.000	1.000	<u>0.752</u>	0.813	1.000	1.000	1.000	0.386	<u>0.853</u>	0.805	0.806	1.000						
15. Fluvia	<u>0.882</u>	-0.013	-0.191	0.076	0.089	-0.191	-0.191	-0.191	<u>0.297</u>	0.531	0.270	0.297	0.568	<u>0.784</u>					
16. Ter	0.720	0.000	0.000	-0.314	-0.200	0.000	0.000	0.000	0.081	0.485	0.111	0.143	1.000	0.860	-0.191				
17. Muga_al	<u>0.887</u>	-0.084	-0.276	0.916	0.912	-0.276	-0.276	-0.276	<u>0.289</u>	0.552	0.326	0.338	0.708	0.860	0.103	-0.276			
18. Muga_Sal	0.903	0.973	0.955	0.976	0.956	0.955	0.955	0.955	<u>0.940</u>	<u>0.980</u>	0.959	0.966	0.935	0.958	<u>0.973</u>	0.955	<u>0.979</u>		
19. Besos	<u>0.780</u>	0.493	0.335	0.634	0.410	0.335	0.335	0.335	0.393	<u>0.560</u>	0.302	0.373	0.333	0.607	<u>0.590</u>	0.335	<u>0.621</u>	0.926	
20. Sovara	<u>0.872</u>	<u>0.938</u>	0.914	0.928	0.907	0.914	0.914	0.914	0.847	0.947	<u>0.912</u>	<u>0.920</u>	0.902	0.926	<u>0.927</u>	0.914	<u>0.943</u>	0.966	0.856

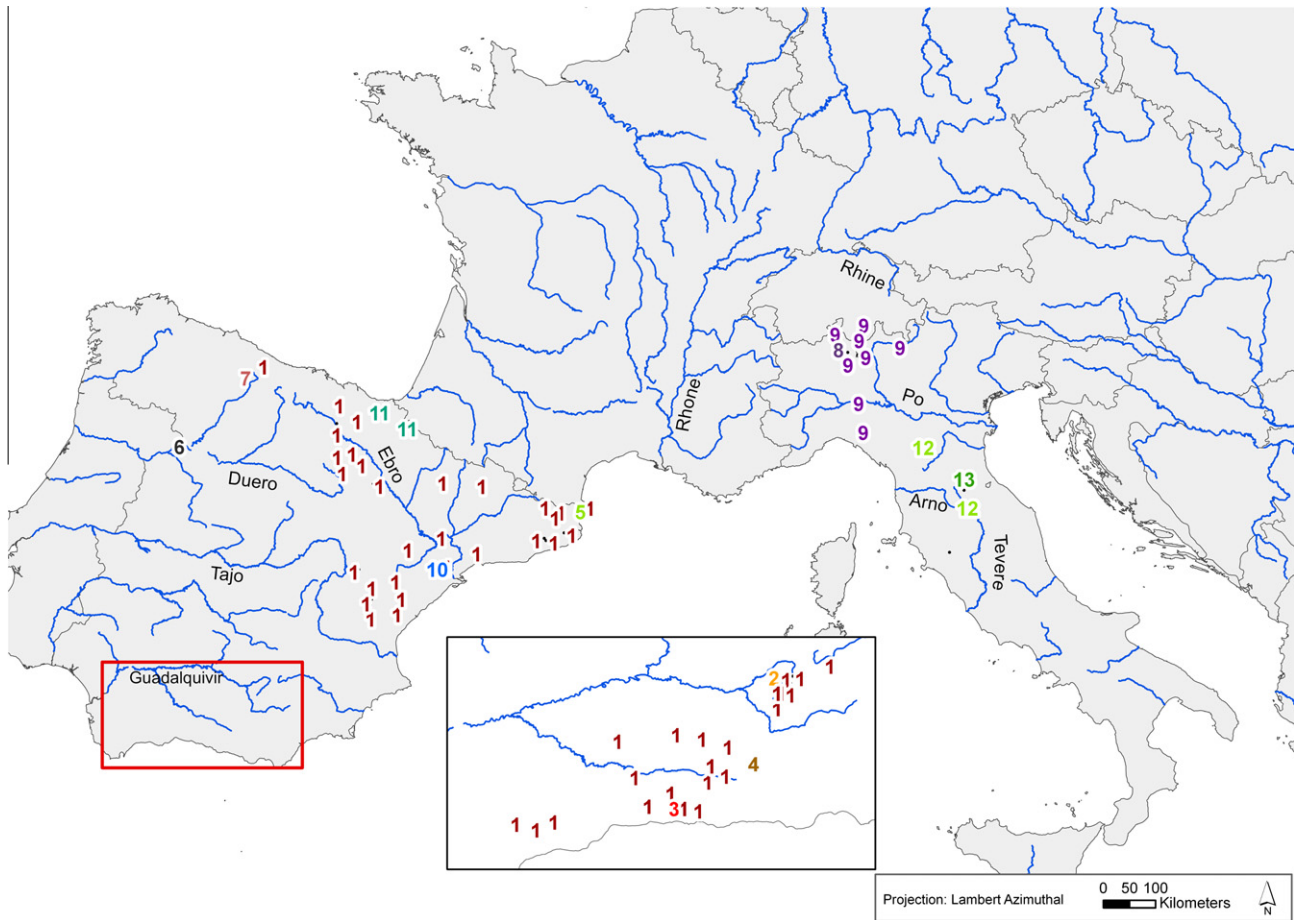


Fig. 4. Map of populations according to significant genetic homogeneous groups of *A. i. italicus* resulting from the SAMOVA analysis (see Table 1 for reference to number codes).

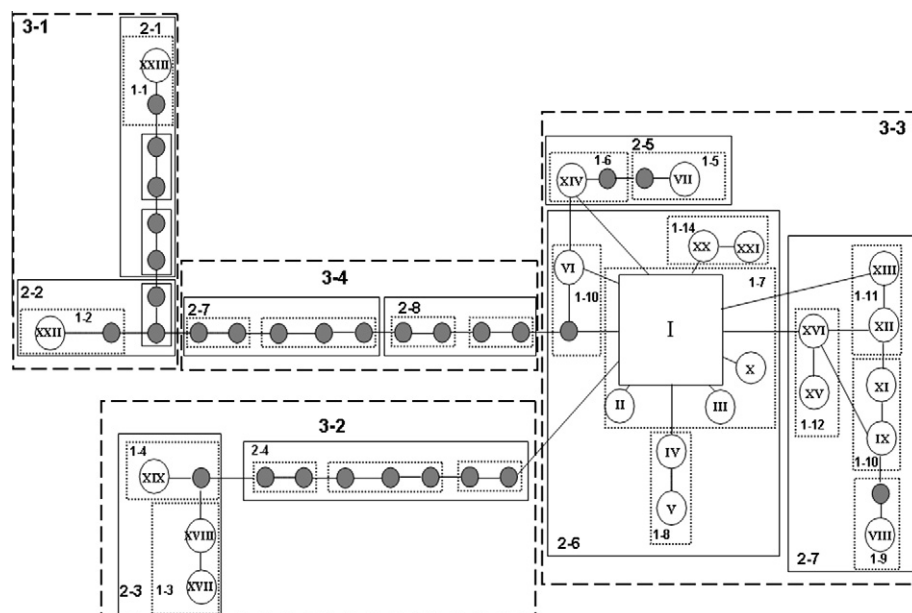


Fig. 5. Statistical parsimony network and nested design (95% probability) of mitochondrial haplotypes of Iberian populations of *A. i. italicus*. Two populations from Italy were included, although main objective of NCA was to test hypothesis of possible geographic patterns on genetic variation of Iberian populations. Small gray circles represent inferred, undetected internal haplotypes, so that each line represents a single nucleotide substitution, dashed line boxes define 1-step clades, square-angled boxes define 2-step clades and thick lines define 3-step clades.

Table 4

Distribution of haplotype frequencies from the NCA at each sampled river basin. See Table 1 for coding of haplotypes and basin assignment for each population.

Region	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	XIII	XIV	XV	XVI	XVII	XVIII	XIX	XX	XXI	XXII	XXIII
Guadajoz (ES)	6																						
Guadalquivir (ES)	42	1	1			1	1							1		2							
Velez (ES)	3																						
Segura (ES)	3																						
Guadalfeo (ES)	3																						
Guadiaro (ES)	5					1																	
Guadalhorce (ES)	3																						
High Ebro (ES)	2			14	3							1			9	2							
Low Ebro (ES)	6															20							
Mijares (ES)	3															3							
Turia (ES)	2															4							
Northern Spain				2	4																		
Fluviá (ES)	10									3			1										
Ter (ES)	3																						
Besós (ES)	2							2	1		1												
Muga (ES)	10																			1	1		
Ebro (ES) (High + Low)	8			14	3							1			9	22							
Independen (ES)t	28					1									7								
East Pyrenees (ES)	25							2	1	3	1		1							1	1		
Arno (IT)																						1	1
Sovara (IT)																	1	2	2				
Total (196)	103	1	1	16	7	2	1	2	1	3	1	1	1	1	9	31	1	2	2	1	1	1	1

Table 5

NCPA inferences as described in the GeoDis key of 11 November 2005. Different steps and conclusions are indicated in the chain-of-inference column.

Clade/haplotype	Chain of inference	χ^2	P	Demographic event inferred
X	1-2-3-5-6-7-8		0.004	Restricted gene flow/dispersal but with some long-distance dispersal over intermediate areas not occupied by the species; or past gene flow followed by extinction of intermediate populations
XV	1-2-3-5-6-7-8		0.000	Restricted gene flow/dispersal but with some long-distance dispersal over intermediate areas not occupied by the species; or past gene flow followed by extinction of intermediate populations
1-7	1-2-3-5-6-7-8	188.63	0.000	Restricted gene flow/dispersal but with some long-distance dispersal over intermediate areas not occupied by the species; or past gene flow followed by extinction of intermediate populations
1-8	1-2-3-5-6-7-8	188.63	0.000	Restricted gene flow/dispersal but with some long-distance dispersal over intermediate areas not occupied by the species; or past gene flow followed by extinction of intermediate populations
1-14	1-2-3-5-6-7-8	188.63	0.023	Restricted gene flow/dispersal but with some long-distance dispersal over intermediate areas not occupied by the species; or past gene flow followed by extinction of intermediate populations
2-6	1-2-3-4 NO	205.7	0.026	Restricted gene flow with isolation by distance
2-7	1-2-3-4 NO	205.7	0.001	Restricted genetic flow with isolation by distance
Total Cladogram	1-19-20 NO	296	0.000	Inadequate geographic sampling

HPD) and 469,613 for the ancestral population ($92,167-7.72 \times 10^6$ 95% HPD). This suggests that current populations in the Iberian Peninsula have larger effective sizes than the ancestral population (congruent with changes in population size estimated by mismatch methods), and are also larger than those of the Italian Peninsula, this result however was not corroborated by the BSP analysis, and should be taken with cautions due to the absence in the analysis of the two populations from France recorded for *A. i. italicus*. Despite of this, we think that the possibility of gene interchange is very low considering that these are two isolated populations and they have been argued to be most probably originated by introduction (Grandjean et al., 2000; Souty-Grosset et al., 2006). As shown by rarefaction analysis, population parameters are probably not strongly influenced by skewed sampling effort between Italian and Iberian groups, mainly due to the high proportion of unique haplotypes from the Iberia Peninsula.

The effective population sizes and demographic trends estimated by the BSP analysis showed a very similar pattern for the Iberian and Italian samples. Only when considering the full data set of Iberian samples (Fig 7A) slightly higher effective population size values were observed and population growth. However, for the reduced and random Iberian data sets, the size and trends of the population showed almost equal values, and overlapping HPDs, than the Italian samples.

4. Discussion

4.1. Phylogenetic relationships of Iberian *Austropotamobius italicus*

This study represents the most comprehensive approach to the phylogeny of the white-clawed crayfish in the Iberian Peninsula, as it covers most of its distribution range in this region. The samples of *A. italicus* used in this study are divided into two main clades (see Fig. 3). According to the phylogenetic tree, most samples from the Iberian Peninsula could be confidently assigned to *A. i. italicus* (Fig. 3, (Fratini et al., 2005; Grandjean and Souty-Grosset, 2000). *Austropotamobius i. carinthiacus* is paraphyletic, with *A. i. italicus* nesting inside it, both conforming Clade I. Clade II was composed of samples previously assigned to sub-species *A. i. meridionalis* and *A. i. carsicus*, however no geographic or sub-specific structure was observed within this clade. The finding of a certain concordance between geography and phylogenetic structure was used to support different sub-species in Clade II. Our findings show that although the two sub-species are reciprocally monophyletic, they are supported by very low bootstrap and posterior probability values. This might suggest that previous studies only included a limited portion of the genetic variability of *A. italicus* (Fratini et al., 2005; Grandjean and Souty-Grosset, 2000). Thus, the inclusion of a larger number of individuals, and therefore larger genetic

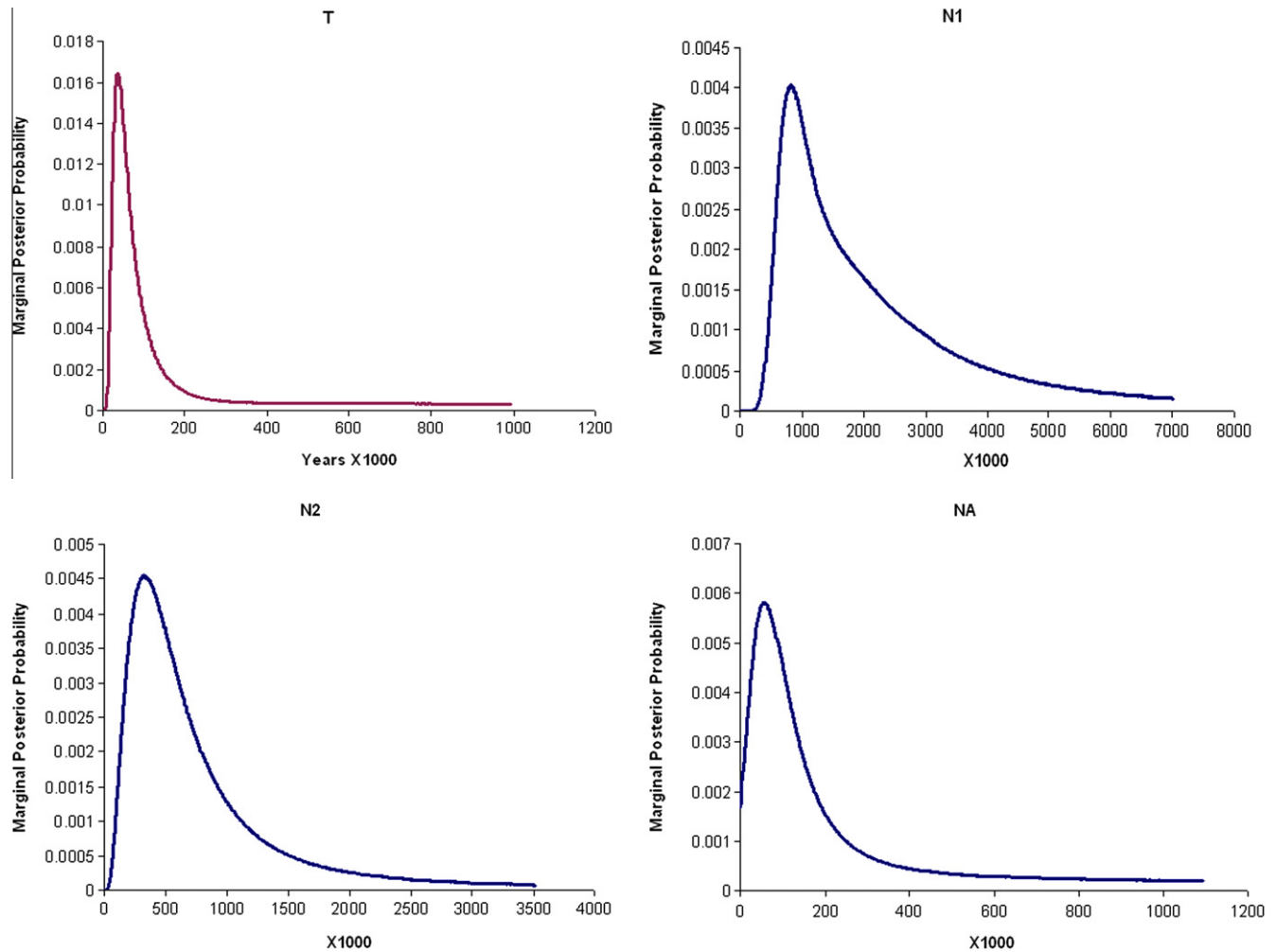


Fig. 6. Probability densities for parameters from the Isolation Migration Model. T is shown on a scale of years over the range corresponding to a maximum t value of 0.16; estimated N ($N = \theta/4(\mu)$) is shown for actual populations of Iberian group (N1), Italian group (N2) and the ancestral (NA) population. Each curve is based upon the results of multiple simulations over millions of Markov chain updates (see Section 2), and is plotted over the specified prior range of each parameter, after standardization.

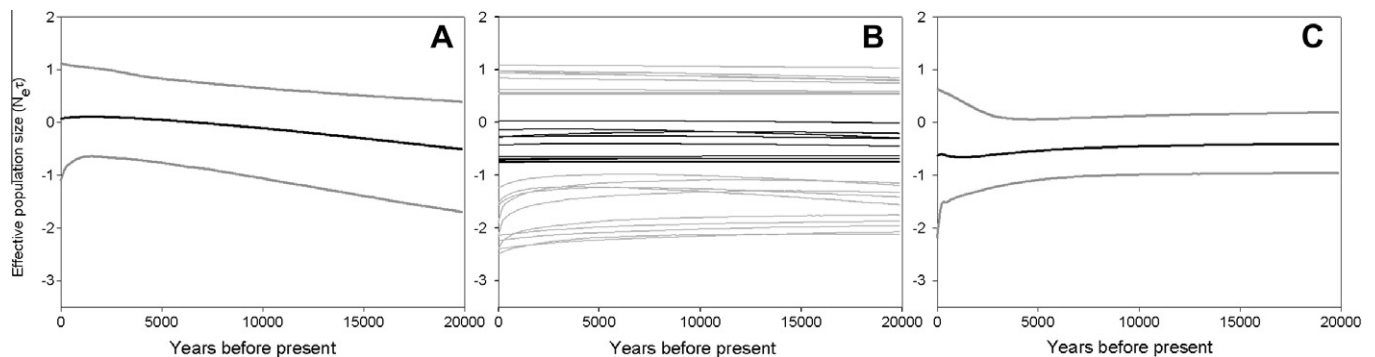


Fig. 7. Bayesian Skyline Plot of Iberian (A) and Italian (B) populations of *A. i. italicus*, showing effective population size over time under neutral evolution. (A) Iberian samples; (B) joint BSP resulting from the random sampling strategy consisted of constructing and analyzing 10 matrices with 20 random Iberian sequences (minimum sample size of the Italian samples); (C) BSP of Italian samples.

variation, could have completed the missing phylogenetic information and could therefore show that the sub-specific genetic differentiation is not so clear.

Overall, Clades I and II could be associated with some degree of geographic structuring according to their distribution in Europe. Thus, Clade I is mainly associated with a western-oriented compo-

nent, including some part of the north Apennines, western Alps, France and the Iberian Peninsula, whereas Clade II has a more eastern distribution that includes the eastern Alps, NE Italy and the Dinaric Alps. Both clades overlap in a large “suture” zone mainly found along the Alpine region. Our phylogenetic evidence therefore suggests a complex and recent biogeographic scenario, thereby

suggesting the complexity of past changes in distributions of the genus *Austropotamobius*. Non-fitting to a general regionalization should not always be considered to be a consequence of human translocation, and any conclusion should be treated with particular caution, especially when based on scarce genetic information. Disjunctive and patchy patterns are frequently observed in different organisms in Europe as a result of climatic change in recent times, especially during glacial periods (Benke et al., 2009; Schmitt et al., 2006; Weiss et al., 2002), and some scarce and rare haplotypes can be explained by historical processes as relicts of previous wider ranges. For example, the higher diversity of haplotypes in the sub-species *A. i. carsicus* and *A. i. carinthiacus* could be identified with a mixed pattern of relict populations associated with the Alpine region, probably related to past glacial microrefugia, as proposed for other regions such as the Istrian Peninsula (Trontelj et al., 2005).

Previous studies with 16S rDNA and COI have found only one haplotype for all the Iberian populations although this observation is not further supported by this study. The 16S rRNA gene showed no difference between all the *A. i. italicus* from Spain, probably due to a slower rate of variation for this marker than COI, as has been observed in previous studies with this species and with other organisms (Trontelj et al., 2005; Verovnik et al., 2005; Zaccara et al., 2004). The finding of a different lineage from the river Muga indicates that genetic diversity in the Iberian Peninsula is not negligible when considering the 16S rDNA fragment. This observation could support alternative hypotheses, in which genetic diversity in the Iberian Peninsula would not occur exclusively due to human introductions (Fratini et al., 2005; Grandjean et al., 2000; Grandjean and Souty-Grosset, 2000; Trontelj et al., 2005). As found in comparative studies with introduced and native populations, an absence of mtDNA differentiation and haplotype diversity would be expected if all populations came from one introduction event and from a very reduced number of populations or even only one population (Miura et al., 2006; Uwai et al., 2006). The evidence that two divergent lineages with contrasting distributions occur in the Iberian Peninsula would require at least two different introduction events to have taken place with different dispersion dynamics subsequent to their introduction. Currently, we have no evidence to suggest that this scenario could have occurred by natural or artificial mechanisms. However, there is not valid argument to favor the alternative hypothesis of at least two human introductions leading to the observed genetic pattern because naturally biogeographic processes are possible and they should be considered alternatively.

4.2. Genetic diversity

Although mitochondrial diversity for the Iberian populations was low, it was higher than reported previously (Grandjean et al., 2002a, 2001; Trontelj et al., 2005). Indeed, several nucleotide diversity values observed for other regions in previous studies of the *A. italicus* complex, were similar to those obtained in this study for the Iberian Peninsula with the same COI fragment ($\pi = 0.001$ and 0.009 for Dalmatia and SE Alps, respectively) (Trontelj et al., 2005). Specifically, this is the case for Clade I, which is characterized by the lowest genetic diversity values of all populations studied to date (Grandjean and Souty-Grosset, 2000; Trontelj et al., 2005). Haplotype divergence is also much lower in this clade than in Clade II. The lower diversity of Iberian *A. i. italicus* could also be due to a strong reduction in population size since the introduction of exotic crayfish such as the red swamp crayfish (*Procambarus clarkii*) and the signal crayfish (*Pacifastacus leniusculus*, (Gherardi and Holdich, 1999)). Both of these species are known to carry the crayfish plague (Diéguez-Urbeondo et al., 1997), which decimated a large part of the remaining populations and would therefore affect the genetic diversity and structure found.

NCA assumes a complete geographic sampling, which is not included in this study due to the lack of possible intermediate populations from France (see Grandjean et al., 2000), possibly this is the reason why only NCA inferences within the Iberian Peninsula were significant. The structure depicted by the parsimony network for Clade I (Fig. 5) shows a well-defined group of haplotypes corresponding to populations of Iberian *A. i. italicus* with very reduced diversity (Clade 3-3). These populations are connected to two clades of Italian samples (Clades 3-1 and 3-2), which presumably pertain to *A. i. carinthiacus* (Population 64) from the river Arno and *A. i. italicus* from the river Tevere (Population 65). The geographic associations between haplotypes from Iberian Peninsula led us to consider this as evidence for the historical presence of *A. i. italicus* in Iberia (Hewitt, 1999). We consider the significant associations involving single-step clades as most probably being due to restricted gene flow, in light of the limitations inherent to the freshwater dispersal of crayfish (Bubb et al., 2008), which could be necessary for long-distance dispersal. However, this kind of movement within Iberia could have been mediated by human translocation. Haplotype analysis showed relatively high haplotype richness for the Iberian Peninsula, and quite different haplotype frequencies between these populations and the Italian (Table 1).

Two haplotypes (Haplotypes 10 and 21) were found to be shared between the Italian and Iberian populations. Although the occurrence of these haplotypes did not show a clear geographical concordance, some patterns could be identified, such as higher haplotype diversities in the NE Iberian Peninsula (the eastern Pyrenees and Ebro basin). These observations are in agreement with recent findings of haplotype and genetic diversity (Beroiz et al., 2008; Dieguez-Urbeondo et al., 2008), which contrast with previous studies with less geographical and genetic sampling from the Iberian Peninsula, where only one mtDNA haplotype had been found (Grandjean et al., 2002a, 2001; Trontelj et al., 2005). Several studies have shown that all haplotypes of introduced populations are represented in the native range of the species, which generally shows higher diversity and effective population sizes (Miura et al., 2006; Uwai et al., 2006). This is not, however, the case for Iberian *A. i. italicus*, as at least 15 COI haplotypes were found exclusively in the Iberian Peninsula. Furthermore, estimates of effective population sizes, a relative measure of genetic diversity, were similar for Iberian and Italian populations, as well as their demographic histories. Despite having high haplotype richness, Iberian populations showed low nucleotide diversity values, which is frequently associated to recent diversification events of population expansion after a period of low effective populations size caused by bottlenecks or founder effects (Slatkin and Hudson, 1991). These findings would, again, not reject the natural origin hypothesis for this species in the Iberian Peninsula.

4.3. Phylogeographic and historical patterns

Historical factors have been shown to affect and shape species distributions (Hewitt, 2004). In some cases, complex phylogeographical patterns can be difficult to explain and are not always coincident with those observed for other organisms. This could be because they represent relict groups from past wider distributions of non-invasive haplotypes after glacial episodes (Gómez-Zurita and Vogler, 2003; Lenk et al., 1999; Schmitt, 2007). Patterns that include the cross-passing of present geographical barriers have been described to explain complex haplotype distribution, as in the case of *Cottus gobio*, a headwater fish species widely distributed in central Europe, whose actual distribution can largely be explained by very recent events associated with consecutive glacial retreat cycles and whose colonization of the Swiss Rhine basin

occurred as recently as after the Younger Dryas, around 6600 years ago (Neuenschwander et al., 2008).

As previously proposed for the white-clawed crayfish, recent climatic factors such as sea-level changes and retreat-expansion cycles of glacial sheets are likely to have played a central role in its diversification and distribution (Dieguez-Urbeondo et al., 2008; Fratini et al., 2005; Santucci et al., 1997). Accordingly, it seems possible that as with other freshwater species, current distribution patterns have been highly influenced by the formation of Pleistocene refugia (Schmitt, 2007), from which posterior dispersion and reinvasion could take place, as proposed by the Expansion–Contraction Model (Hewitt, 1996; Provan and Bennett, 2008). In addition, mitochondrial differentiation of the Iberian populations is supported by the Φ_{ST} and SAMOVA analyses, which clearly indicated a genetic structure between the Iberian and Italian Peninsula, indicating that these populations could have a natural origin. Within the Iberian Peninsula most populations seem to constitute a large genetic group, suggesting low differentiation or high gene flow, either natural or human-mediated.

Genetic variation in Iberian populations of *A. i. italicus* could also be linked to demographic responses to the retreat and recovery of the ice during and after the last glacial maxima (LGM). Considering the hypothesis first proposed by Karaman (Karaman, 1962) and supported further by Santucci et al. (1997) and Dieguez-Urbeondo et al. (2008), it is possible that the sub-species *A. i. italicus* had a unique distribution area ranging from the Iberian Peninsula eastwards until at least the Central Apennines. This distribution range should be present before the LGM, around 30,000 BP (Hewitt, 2004). According to the results obtained from the IM analysis, splitting could have taken place during the early LGM ($T = 34,858$ years, 90% HPD = $20,261 - 1.501 \times 10^6$ years), when temperatures dropped in Northern Europe but, which is probably more important for freshwater organisms, a major desiccation process occurred over most of the territory between the Pyrenees and the Alps (Peyron et al., 1998; Wu et al., 2007). It is possible that this process separated the population history of both Iberian and Italian populations of this species. After the LGM, Iberian populations could have been strongly reduced to only a few populations, that expanded their geographic range to the rest of the Iberian Peninsula, around 10,772–20,004 years BP, as suggested by the time since expansion estimated by the coalescent simulations performed in the mismatch analysis (Supplementary material II). However, this range expansion was not necessarily related to an increase in the effective population size, neither in the Iberian or Italian Peninsula, as indicated by the BSP (Fig. 7).

These dates are also in accordance with the splitting times estimated for the black pine *Pinus nigra*, from the Pyrenees and Southern France, which presumably separated just before the LGM, possibly as a consequence of a higher desiccation rate in the eastern region of the Pyrenees (Afzal-Rafii and Dodd, 2007). It seems difficult however for *A. i. italicus* to reach almost all river basins in the Iberian Peninsula due to its low dispersal capability (Bubb et al., 2008). Furthermore, considering that the first direct historical record for the species dates from the 18th century (Alonso, 2004), it is possible that, like many other species of economic and/or conservation interest, recent human-mediated movements could have played an important role in its current distribution within the Iberian Peninsula (Muñoz, 1967), which could be related to the low genetic differentiation observed among localities. Glacial patterns have also been proposed to shape *A. italicus* distribution in the Italian and Istrian Peninsulas, where haplotype ranges sometimes show patchy and discontinuous patterns (Trontelj et al., 2005). Several species and haplotypes have similar disjunctive distributions between the Iberian and Italian populations, mainly due to glacial processes. This is also the case for a number of species with puzzling distributions, such as the

populations of *Salamandra* conformed by close haplotypes of *S. s. bernardezi* found in Northern Spain and *S. s. giglioli* from Southern Italy (Steinfartz et al., 2000); several mountain butterflies found in the Pyrenees and the SW Alps (Schmitt et al., 2006); the bushcricket *Ephippiger ephippiger* in the Pyrenees and the Alps (Spooner and Ritchie, 2006); the chamois *Rupicapra pyrenaica* (Masini and Lovari, 1988) with populations distributed in the Pyrenees and Southern Italy; the European hares *Lepus castroviejoi* and *L. corsicanus* (Alves et al., 2008) being two very closely-related species occurring in Northern Spain and Southern Italy, Sicily and Corsica; and even plant species like the pubescent oak *Quercus pubescens* (Petit et al., 2003), which is found in the central Apennines and Southern France, and as far as the Pyrenees. Several explanations have been argued for these allopatric distributions of closely related groups, generally including a very recent time of splitting during the Würm glacial period and different post-glacial dispersion dynamics between different lineages, which could have led to the patchy pattern observed today.

The mitochondrial evidence presented here provides new arguments to consider a possible natural origin for the Iberian populations of *A. italicus* as a consequence of a complex biogeographic and demographic history and further recent natural or human-mediated processes that might have shaped its current genetic structure. Future studies should involve a more exhaustive sampling effort from Italy and consider the nuclear variation in a multi-loci framework to help define the time and mechanisms subjacent to the biogeography of this sub-species.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.ympev.2010.06.007.

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